

Monaural Speech Enhancement: From Shallow Networks to Speech Foundation Models

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**Università
degli Studi
di Palermo**

Outline

Introduction

Early Attempts

Deep Learning Era: Denoising Auto Encoders

Deep Learning Era: Mapping Approach

Deep Learning Era: Masking Approach

Deep Learning Era: Architectural Evolution

Deep Learning Era: Time Domain Approach

Deep Learning Era: Generative Approach

Deep Learning Era: Vocoder-based Approach

Deep Learning Era: Foundational Model

Open Challenges & Conclusions

References

Outline

Introduction

Early Attempts

Deep Learning Era: Denoising Auto Encoders

Deep Learning Era: Mapping Approach

Deep Learning Era: Masking Approach

Deep Learning Era: Architectural Evolution

Deep Learning Era: Time Domain Approach

Deep Learning Era: Generative Approach

Deep Learning Era: Vocoder-based Approach

Deep Learning Era: Foundational Model

Open Challenges & Conclusions

References

Problem Statement: Why Speech Enhancement?

“Can you hear me now?”

A question we've all asked on a noisy call, in a crowded room, through a hearing aid.

Speech enhancement (SE) is the invisible front-end that answers it - Making speech clearer wherever noise gets in the way: Telecommunications, hearing aids, voice-controlled systems.



Quality

Does it sound *natural*? Comfortable to listen to, without fatigue.



Intelligibility

Can you *understand* it? Every word, clearly and correctly.

Neural Speech Enhancement: Scope & Roadmap

This talk surveys **single-microphone** speech enhancement.

Most desirable from an application standpoint (single microphone)

Less sensitive to **room reverberation** and **spatial source configuration** vs. multi-mic systems

We trace the evolution of neural speech enhancement **from simple shallow models to modern foundational models**:

Problem Setup for Speech Enhancement

SE can be tackled in either the time domain or the frequency domain.

Time-Domain Formulation

observed noisy signal clean speech additive noise

↓ ↓ ↓

$$y(n) = x(n) + d(n)$$

Frequency-Domain Formulation (via DTFT)

$$Y(\omega) = X(\omega) + D(\omega)$$

Polar Representation of $Y(\omega)$

magnitude spectrum

↓

$$Y(\omega) = |Y(\omega)| e^{j\phi_y(\omega)}$$

↑

phase spectrum

Most classical and early neural methods estimate $|X(\omega)|$ and reuse $\phi_y(\omega)$.

Outline

Introduction

Early Attempts

Deep Learning Era: Denoising Auto Encoders

Deep Learning Era: Mapping Approach

Deep Learning Era: Masking Approach

Deep Learning Era: Architectural Evolution

Deep Learning Era: Time Domain Approach

Deep Learning Era: Generative Approach

Deep Learning Era: Vocoder-based Approach

Deep Learning Era: Foundational Model

Open Challenges & Conclusions

References

Early Neural SE: Feedforward Networks

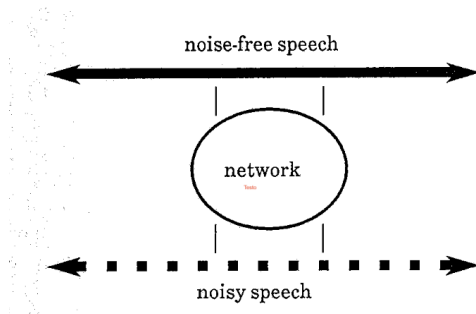
Feed-forward multi-layer neural networks were among the earliest neural approaches to speech enhancement, dating back to the **late 80s** with the work by Tamura & Waibel [1].

Architecture: 4-layer feedforward network, 60 sigmoidal units per layer

Input: raw waveform, 60 samples per frame

Data: 5,000 isolated Japanese words + artificial room noise at ≈ 20 dB SNR

Training: 5 days



Early Neural SE: Recurrent Networks

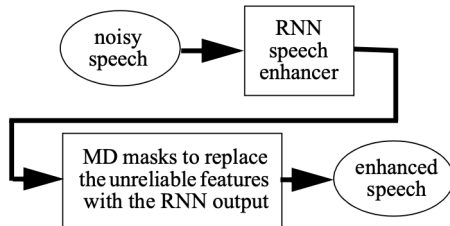
RNNs followed, exploiting temporal context to map noisy speech directly to clean speech in Parveen & P. Green's work [2].

Elman RNN: noisy $\rightarrow$ clean spectra,
direct mapping

I/O: 32-channel spectra, 30 hidden
units, BPTT

Data: AURORA digits, 4 noise types,
20–0 dB SNR

Loss: Sum of Squared Errors (SSE)
(estimated vs. clean spectra)



Outline

Introduction

Early Attempts

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Deep Learning Era: Architectural Evolution

Deep Learning Era: Time Domain Approach

Deep Learning Era: Generative Approach

Deep Learning Era: Vocoder-based Approach

Deep Learning Era: Foundational Model

Open Challenges & Conclusions

References

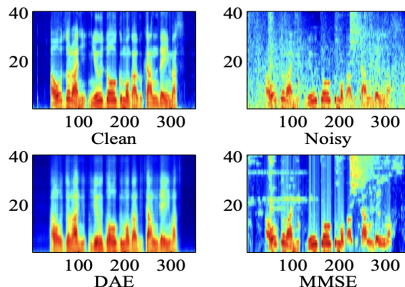
Speech Enhancement Based on Deep Denoising Autoencoder [3, 4]

National Institute of Information and
Communications Technology and Academic
Sinica

First stage: DAE trained on clean Log Power
Spectrum (LPS) representations only:
Reconstruction,

Second stage: Denoising via fine-tuning
DAE explicitly learns the **statistical difference**
between clean and noisy speech

Compared against: MMSE + improved
minimum controlled recursive averaging (IMCRA)
noise tracking



DAE: less severe **speech distortion** and less
noise residue. Tested on **factory** and **car**
noise.

Outline

Introduction

Early Attempts

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Deep Learning Era: Masking Approach

Deep Learning Era: Architectural Evolution

Deep Learning Era: Time Domain Approach

Deep Learning Era: Generative Approach

Deep Learning Era: Vocoder-based Approach

Deep Learning Era: Foundational Model

Open Challenges & Conclusions

References

The Breakthrough: DNN-Based Speech Enhancement (Xu et al., 2013/2015) [5][6]

“A Regression Approach to Speech Enhancement Based on Deep Neural Networks”

Yong Xu, Jun Du, Li-Rong Dai, Chin-Hui Lee

2018 IEEE SPS Best Paper Award

Framework: Regression-based SE using **multi-layer DNNs**, mapping noisy speech directly to clean speech - **Mapping based Speech Enhancement**

Noise reduction: Avoids the **musical artifacts** plaguing classical DSP-based solutions

Generalization: Strong performance on **mismatched, multi-condition** test environments, trained on **2500 hours** of simulated data

Impact: Hundreds of citations - a foundational baseline for modern real-time SE and ASR pipelines

Mapping-Based Deep Deeply Connected Neural Networks

Key idea: Use a **large training set** to give the network enough modeling capacity

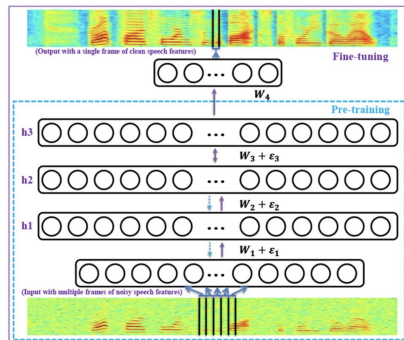
Paired clean-noisy waveforms are needed for training.

Generated artificially from a *clean speech* dataset (**TIMIT**) and a *noise* dataset (**NOISEX-92**).

TIMIT utterances corrupted with **104 noise types** at six SNR levels (20 to -5 dB) $\rightarrow$ **multi-condition training set**

Result: $\approx$ **2500 hours** of noisy training data (plus one clean-condition set)

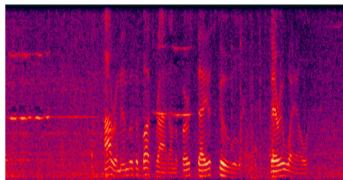
The **MMSE criterion** in the **log-power spectral (LPS)** domain for training



Mapping-based DNN: learning a nonlinear mapping from noisy to clean speech.

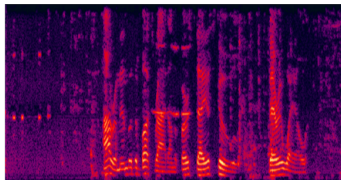
Listen & Compare: Enhanced Results on Real-World Speech

Spectrograms of an utterance extracted from the movie Forrest Gump: Noisy (left), Log-MMSE (middle), and DNN (right) with unseen noise.



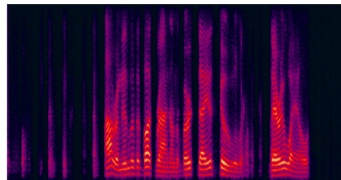
Noisy speech

► click to play



DSP (MMSE) enhanced

► click to play

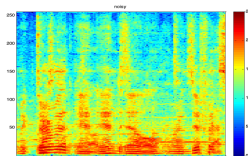


DNN enhanced

► click to play

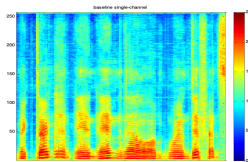
Extension to Multi-Channel SE & Dereverberation [7, 8, 9]

Noisy & reverberant (PESQ: 2.24)



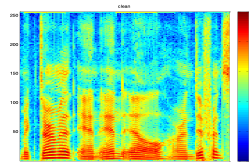
► click to play

Single-channel (PESQ: 2.45)



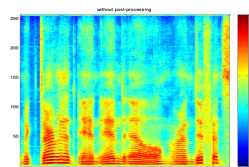
► click to play

Clean (PESQ: 4.0)



► click to play

Multi-channel (8-ch) (PESQ: 3.10)



► click to play

Theoretical Foundations of Mapping-Based Solutions [10, 11, 12]

Mapping-based solutions rest on **extended representation and approximation theorems**, notably the **Kolmogorov–Arnold theorem** and **Cybenko’s theorem** — the theoretical basis for neural networks’ ability to approximate *complex nonlinear mappings*.

Extensions & recent advances

Qi, Siniscalchi, Du & Lee – T-ASLP, 2019	Extends universal approximation theory
Qi, Du, Siniscalchi, Ma & Lee – T-SP, 2020	Derives an upper bound on generalization error
Qi, Du, Siniscalchi, Ma & Lee – SPL, 2020	MAE analysis for DNN-based vector-to-vector regression — <i>Lipschitz continuity</i>

IEEE SPS Best Paper Award 2025

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Introduction

Early Attempts

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Deep Learning Era: Foundational Model

Open Challenges & Conclusions

References

Masking based Speech Enhancement - Prof. Wang's team

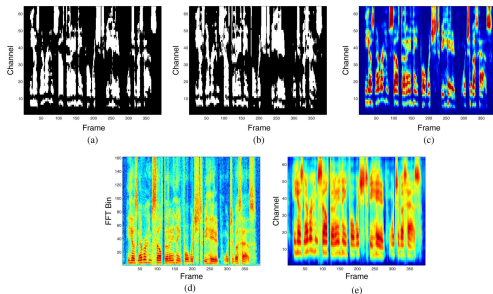
Core Idea [13, 14]

Formulate speech separation as a **super-vised learning** problem, specifically a **bi-nary classification** problem

Neural model learns a mapping from noisy features to a **time-frequency mask**

Training Targets Compared

- (a) Ideal Binary Mask (**IBM**)
- (b) Target Binary Mask
- (c) Ideal Ratio Mask (**IRM**)
- (d) FFT Magnitude & Mask (**FFT-MASK**)
- (e) Gammatone Power Spectrum



Key Findings

Ratio masks (IRM, FFT-MASK) outperform other targets

Outperforms **NMF** and traditional SE methods

Complex Ratio Masking for Monaural Speech Separation [15]

Limitation of Prior Work

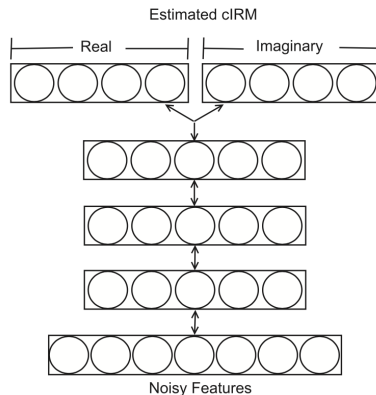
Standard STFT-based systems enhance only the **magnitude** spectrum

Phase spectrum is left unchanged

Proposed Approach

The complex-valued spectrogram is decomposed into real and imaginary part in the Cartesian coordinate

DNN estimates the **real** and **imaginary** components of the **complex ideal ratio mask (cIRM)**



Note: NN remains real-valued.

Comprehensive Overview [16]

A Landmark Survey

“Supervised Speech Separation Based on Deep Learning: An Overview” [16] — the key reference unifying supervised speech enhancement and separation

Topics Covered

- Spectral masking
- Complex-domain mapping
- Training targets

Masking vs. Mapping



Key Insight

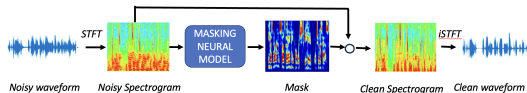
Masking excels at **higher SNRs**, while **mapping** is more advantageous at **lower SNRs**

Signal Approximation [17]

Key Idea

Train a **ratio mask** estimator that minimizes the difference between clean and estimated speech magnitudes

SA combines **ratio masking** and **spectral mapping**, aiming to maximize SNR



Loss function:

$$SA(t, f) = [RM(t, f) |Y(t, f)| - |S(t, f)|]^2$$

Outline

Introduction

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Deep Learning Era: Masking Approach

Deep Learning Era: Architectural Evolution

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Open Challenges & Conclusions

References

Evolution of Neural Blocks in Speech Enhancement

Mapping and masking told the network what to predict.

The next step was designing architectures that better capture the structure of speech itself.



Convolution

Local time-freq. patterns



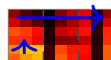
Recurrent

Dynamic frame correlations



Encoder-Decoder

Encoder gives abstract features



Intra/Inter-chunk

Local + global chunk corr.



Attention (chunk)

Same idea via self-attention



Sub-band modeling

Freq.-wise stationarity



Complex operator

Models Re & Im jointly



Multi-path

Separate mag./phase streams



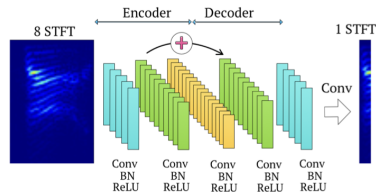
State-space (Mamba)

Redundant Convolutional Encoder-Decoder (CED) [18]

First encoder-decoder **CNN** architecture in T-F domain for SE.

Encoder-decoder learns a **better feature representation** of the noisy spectrum before projecting it back to clean.

The **decoder** gradually compresses back down to the clean spectrum.



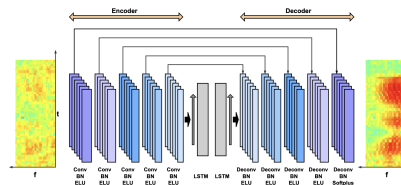
CRN: Convolutional Recurrent Network for Real-Time SE [20]

CRN combines a **convolutional encoder-decoder (CED)** with **recurrent layers (LSTM)**, jointly modeling local spectro-temporal structure and long-term dependencies from magnitude spectra.

Why add recurrence?

Tracking a speaker needs **long-term context** - CED alone **cannot capture** this.

Experimental results showed superior performance compared to fully LSTM-based SE methods [19].



DCCRN: Deep Complex Convolution Recurrent Network for Phase-Aware Speech Enhancement [23]

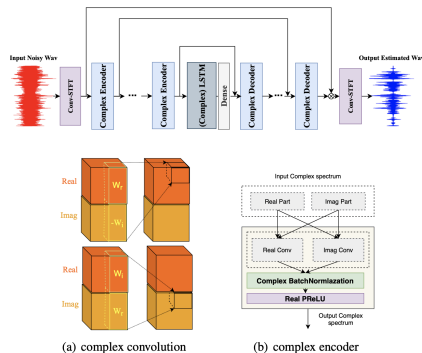
Introduces **complex-valued** convolutional & recurrent modules to model real/imaginary components directly

vs. prior work [15, 21, 22]: earlier methods used *real-valued networks* on complex targets

Result: 1st in MOS evaluation - Interspeech 2020 DNS challenge real-time-track.

Interlude

The complex Conv2D block was based on DCUnet, withdrawn from ICLR 2019 due to issues with the reported results.



Subband-based modeling for SE [24, 25]

Core idea

One **tiny LSTM**, shared across all frequencies, applied independently per subband – *not* one big network on the full spectrum.

Output: cIRM mask per frequency $\rightarrow$ clean speech.

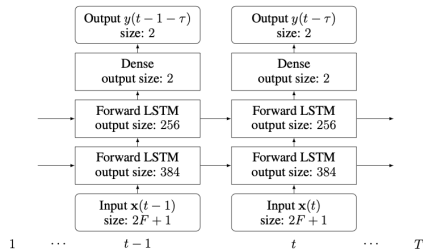
Why it works

Speech vs. stationary noise: seen in the temporal pattern of a **single frequency bin**.

Non-stationary noise: neighboring bins add **local spectral pattern**.

Extension

FullSubNet: combines Full- and Sub-band approaches – outperforming competing models on the DNS Challenge (INTERSPEECH 2020)



PHASEN: A Phase-and-Harmonics-Aware Speech Enhancement Network [26]

Two-path network: Amplitude stream + Phase stream, exchanging information between paths.^a

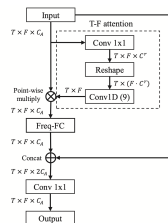
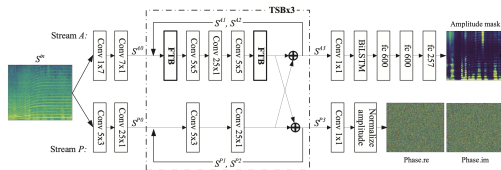
Why? Estimated imaginary cRM $\approx 0 \Rightarrow$ degenerates to **IRM**.

Conv2D limitation: local-only correlations — misses distant **harmonics**.

FTB $\rightarrow$ models **global** frequency correlations through attention.

Attention built using 1×1 Conv + Conv1D + matrix-mult frequency block

^aInput: Magnitude+Phase.



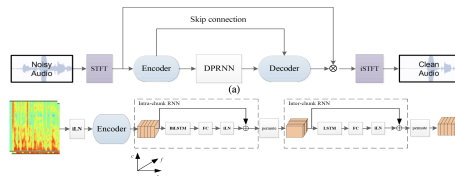
DPCRNN: Dual-Path Convolution Recurrent Network for SE [28]

Key idea: Inspired by DPRNN [27], & adapted to T-F domain

Intra-chunk RNN: captures local spectral patterns within each frame.

Inter-chunk RNN: models dependencies across consecutive frames.

Good results on the wideband track of the Interspeech 2021 DNS Challenge.



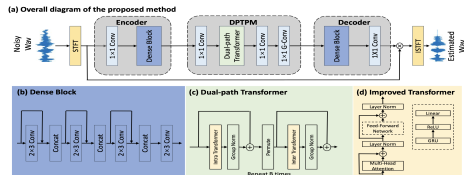
DPT-FSNet: Dual-Path Transformer Full/Sub-Band Network [29]

DPT-FSNet idea

A convolutional encoder–decoder learns a compact latent representation, while a **dual-path Transformer** alternates:

- **Intra-Transformer:** models **sub-band** dependencies.
- **Inter-Transformer:** integrates sub-bands to capture **full-band** information.

Bigger picture: jointly modeling *local frequency structure* and *global spectral context* improves speech enhancement while benefiting from the long-range modeling capability of Transformers.



CMGAN: Conformer-based Metric GAN for SE [31]

Multi-branch complex representation learning^a

Joint processing of magnitude, real, and imaginary spectrograms with dedicated decoders for complementary enhancement.

Dilated DenseNet encoder

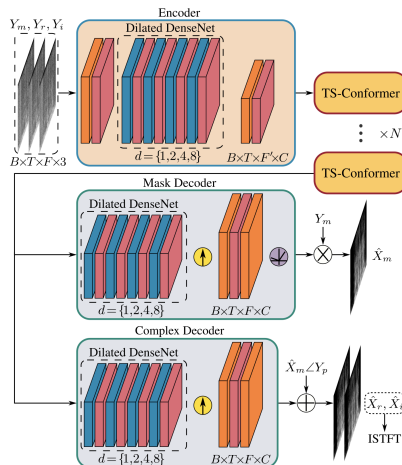
Learns compact and expressive representations from complex spectral inputs.

Two-stage Conformer

Captures temporal dependencies followed by frequency correlations.

Metric-aware GAN training

Combines reconstruction objectives with adversarial learning to improve perceptual quality.



^aD2Former [30] similar with complex neural modules

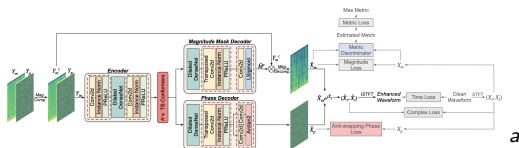
MP-SENet: Parallel Denoising of Mag. and Phase Spectra [34]

Codec-style architecture: Encoder & decoder connected through **convolution-augmented Transformers**.

Two-stage Conformer (TS-Conformer): captures **temporal** and **spectral** dependencies before parallel decoding.

Parallel magnitude & phase enhancement: separate decoder branches estimate magnitude and phase spectra

Multi-objective training: combines metric, time-domain, complex spectral, and anti-wrapping phase losses.



VoiceBank+DEMAND (2023): best reported PESQ = 3.50, STOI = 0.96.

^aThis idea is furthered by ZipEnhancer [32], and BSMP-SENet[33]

Mamba: A novel state-space-model for SE

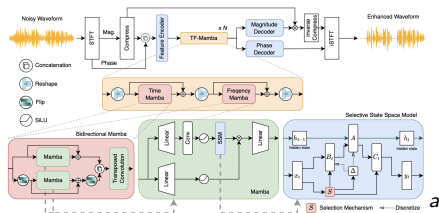
Exploits **Attention-free** solutions to capture **long sequences** while reducing model complexity.

[35]: **MP-SENet** framework with Mamba insted of Conformer - top results & **~12% FLOPs reduction**.

[36]: **U-Net** structure (as in [37]) in a **multi-path** framework.

[38]: **Band-split** framework [39] - but full-band split is non-uniform and processed by Mamba.

[40]: RT-SEMamba Real-Time SE via Progressive Knowledge Distillation.^a



^aConformer remarkable performance but quadratic computational complexity and high hardware requirements. Mamba replaces Conformer.

^aTuesday 29 September 2026 - 16:30 - Session
Real-Time, Low-Latency and Edge Speech Enhancement

Outline

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Deep Learning Era: Masking Approach

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Deep Learning Era: Generative Approach

Deep Learning Era: Vocoder-based Approach

Deep Learning Era: Foundational Model

Open Challenges & Conclusions

References

Time-Based Speech Enhancement

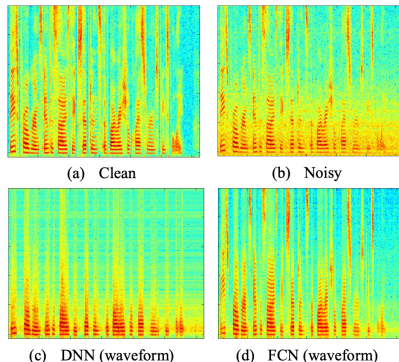
Time-domain SE avoids **explicit phase estimation** and T-F reconstruction inconsistencies - reducing artifacts.

Fu et al. (2017) [41] time-domain modeling:

Fully connected layers struggle to generate high- and low-frequency content **jointly**, despite being universal approximators in theory.

Fully convolutional layers model both **simultaneously** - confirmed by spectrogram evidence.

L2 norm based loss used in training.



DEMUCS for SE [43]

Conv encoder-decoder + U-Net skip connections, with a **sequence model** on the encoder output

Layers indexed 1 to L (decoder reversed) $\Rightarrow$ matching scales share an index

Pipeline

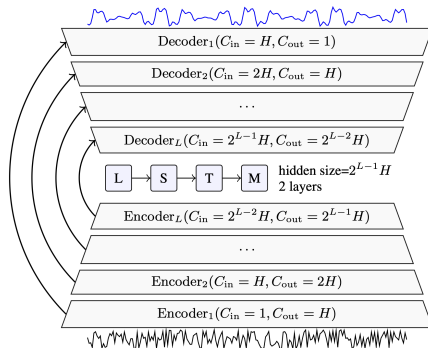
Encoder: $E(x) = z$

Sequence model: $LSTM(z) + z = \hat{z}$

Decoder: $D(\hat{z}) = \hat{y}$

Skip connections: encoder layer $i \rightarrow$ decoder layer i

In 2020: outperformed **SEGAN**, **MetricGAN**



DEMUCS adapts Conv-Tasnet architecture [42]

Outline

Introduction

Early Attempts

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Deep Learning Era: Time Domain Approach

Deep Learning Era: Generative Approach

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Open Challenges & Conclusions

References

SEGAN: Speech Enhancement Generative Adversarial Network [44]

Generator (G): Fully encoder-decoder structure

Input signal compressed into a **latent representation** (thought vector c)

c combined with a latent vector z , sampled from a **standard Gaussian**

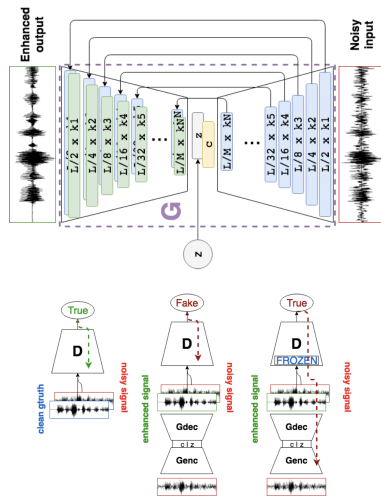
Training pairs for Discriminator (D):

Real: $(\tilde{x}, x)$ — noisy, clean

Fake: $(\tilde{x}, \hat{x})$ — noisy, enhanced

Data: Waveform chunks ≈ 1 s (16384 samples), sliding window with **50% overlap**

Inference: non-overlapping windows, outputs **concatenated** to reconstruct the full signal



MetricGAN: GAN-Based Black-Box Metric Optimization for SE [50]

Several improvements over SEGAN followed over the years [45, 46, 47, 48].

MetricGAN: Instead of a fixed false label (“0”) for generated speech, the discriminator is trained with **labels given by the evaluation metric** itself, e.g., PESQ, or STOI.

	PESQ	CSIG	CBAK	COVL
Noisy	1.97	3.35	2.44	2.63
SEGAN	2.16	3.48	2.94	2.80
MMSE-GAN	2.53	3.80	3.12	3.14
WGAN-GP	2.54	-	-	-
Deep Feature Loss	-	3.86	3.33	3.22
SERGAN	2.62	-	-	-
MetricGAN (P)	2.86	3.99	3.18	3.42

The discriminator’s target space shifts from **discrete** (real/fake) to **continuous** (evaluation score).^a

^aIn 2023, GAN-in-GAN [49] further MetricGAN: (i) Inner GAN for spec-to-spec recovery, (ii) Outer GAN for audio-to-audio recovery. Competitive with CMGAN.

VAE-Based Probabilistic SE

*VAE-based probabilistic formulation enables **robustness under uncertainty**.*

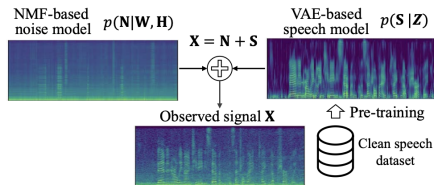
Early work [51, 52, 53, 54]:

VAE learns a **prior distribution** of clean speech - $p(\mathbf{X}|\mathbf{Z})$

Combined with an **untrained NMF** noise model - $p(\mathbf{N}|\mathbf{W}, \mathbf{H})$

Signal variances estimated via **Monte Carlo expectation maximization (MCEM)**

Enhancement: Masking of the mixture signal, akin to probabilistic **Wiener filtering**.



Disentangling Speech and Noise with independent VAEs [54]

VAE-based methods cannot disentangle speech and noise latent variables, so speech is modeled with a VAE while noise uses a traditional NMF model.

Solution: disentanglement via explicit, independent modeling:

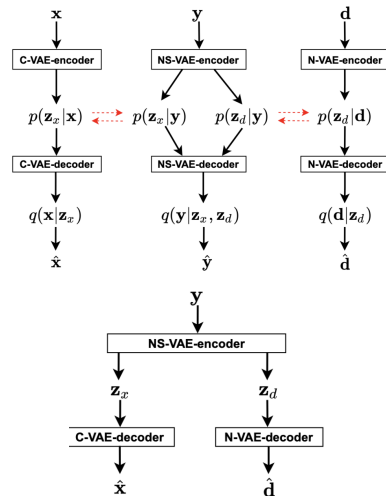
C-VAE trained on clean speech only

N-VAE trained on noise only

NS-VAE trained on noisy speech, leveraging the C-VAE and N-VAE encoders

At inference, the NS-VAE encoder takes the noisy speech and produces $\mathbf{z}_x$ and $\mathbf{z}_d$, sent to the C-VAE and N-VAE decoders, respectively.

Result: disentanglement confirmed on simulated experiments.

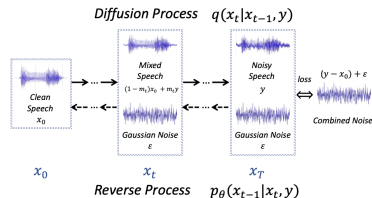


Diffusion Models for SE in Time Domain

Diffusion models were effectively deployed for SE by incorporating the **observed noisy speech** directly into the **forward and reverse processes** [56, 57].

Inference: Starts from the **noisy observation** corrupted by isotropic Gaussian noise.

Reported results, in 2022, better than competitive time-domain generative models on VoiceBank.



Diffusion process:

$$q_{\text{diff}}(x_t | x_0, y) = \mathcal{N}\left(x_t | (1 - m_t) \sqrt{\bar{\alpha}_t} x_0 + m_t \sqrt{\bar{\alpha}_t} y, \delta_t \mathbf{I}\right)$$

where $m_0 = 0$ and $m_T \simeq 1$

Reverse process:

$$p_{\text{diff}}(x_{t-1} | x_t, y) = \mathcal{N}\left(x_{t-1} | \mu_{\theta}(x_t, y), \tilde{\delta}_t \mathbf{I}\right)$$

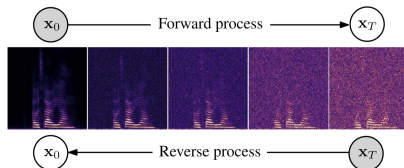
Score-Based Generative Models for SE [58, 59]

Diffusion process based on a **continuous-time stochastic differential equation (SDE)**, with a **variance-exploding (VEDM)** formulation.

Operates on **complex STFT domain**.

Forward process: Moves from clean to noisy speech via a **drift term** $\Rightarrow$ purely generative objective.

Reverse process: Moves from a mixture of **noisy speech and Gaussian noise** back toward clean speech.



Forward process:

$$d\mathbf{x}_t = \underbrace{\mathbf{f}(\mathbf{x}_t, \mathbf{y}) dt}_{\text{drift term}} + \underbrace{g(t) d\mathbf{w}}_{\text{diffusion term}}$$

$$\mathbf{f}(\mathbf{x}_t, \mathbf{y}) = \gamma(\mathbf{y} - \mathbf{x}_t)$$

Reverse process:

$$d\mathbf{x}_t = \left[-\mathbf{f}(\mathbf{x}_t, \mathbf{y}) + g(t)^2 \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t | \mathbf{y}) \right] dt + g(t) d\hat{\mathbf{w}}$$

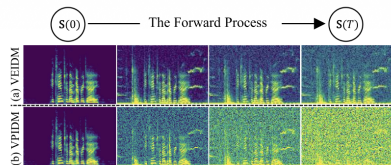
$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t | \mathbf{y})$: the *score* — direction of steepest increase of $p(\mathbf{x})$, estimated by a neural network.

Variance-Preserving Interpolation Diffusion Model for SE [62, 63]

VEIDM (variance-exploding): needs a **corrector step**, doubling inference cost; its initial state is only **approximated** from noisy speech, causing errors that propagate through the reverse process.

VPIDM (variance-preserving): shrinks the speech component as $\tau \rightarrow T$, so the **terminal state** is dominated by Gaussian noise — unlike VEIDM's full-scale signal.

Result: smaller initial error and more accurate noise estimates **eliminate the corrector step**, cutting sampling steps from ~ 60 (VEIDM) to **25** (VPIDM), with better performance at **low SNR**.



Hallucinations in adverse scenarios have been observed^a - Bridge models have been proposed to solve it [60]

Gaussian noise schedule & loss functions impacts the DM performance $\rightarrow$ A new DM loss in [61]

^aVocalization or breathing artifacts in low SNR or in speech absence

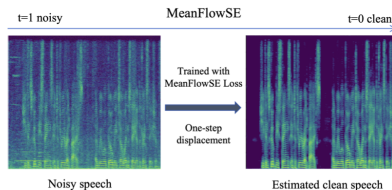
Flow Matching for SE [64, 65]

Models SE as **continuous probability transport** between noisy and clean speech distributions.

Learns transport **vector fields** directly, typically as **ordinary differential equations (ODEs)** — no iterative stochastic denoising.

vs. Diffusion: far **higher sampling efficiency**, fewer function evaluations, competitive quality.

Appeal: well suited for **real-time** and **resource-constrained** applications.



Improving Generative Models for SE

To address **hallucinations** and/or **computational inefficiency**, several works combine **deterministic** and **diffusion-based** SE solutions.

Two integration strategies: **independently trained** blocks, or **jointly trained** systems.

StoRM [66]: The **first** work in this direction; components are **disjointly trained**.

Thunder [67]: Links regression and diffusion SE via a **Brownian bridge**.

SIPS [68] (*very recent*): Uses a **predictive–generative drift decomposition** to improve the combination.

Today's section: one-step denoising solutions:

Speech Enhancement Based on **Drifting Models** for one-step noise removal

Schrödinger Bridge Mamba for one-step speech enhancement

Learnable Schrödinger Bridge and Activations for efficient diffusion-based speech enhancement

Outline

Introduction

Early Attempts

Deep Learning Era: Denoising Auto Encoders

Deep Learning Era: Mapping Approach

Deep Learning Era: Masking Approach

Deep Learning Era: Architectural Evolution

Deep Learning Era: Time Domain Approach

Deep Learning Era: Generative Approach

Deep Learning Era: Vocoder-based Approach

Deep Learning Era: Foundational Model

Open Challenges & Conclusions

References

Speech Enhancement with Neural Vocoders

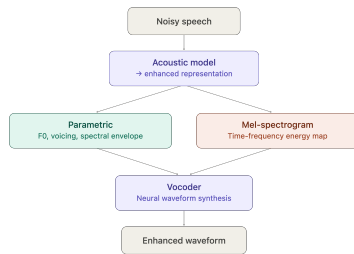
Key idea: decouple enhancement from waveform generation via an intermediate **acoustic representation**.

Enhancement network predicts a **clean representation** from noisy speech

A **neural vocoder** generates the waveform from it

Effectiveness is **tightly coupled** to progress in **vocoder technology**.

Evolution: Initial work by Maiti & Mandel [69]



Main limitations:

Strong **out-of-distribution** speaker generalization not demonstrated

Risk of **content/phoneme hallucination**

Outline

Introduction

Early Attempts

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Deep Learning Era: Mapping Approach

Deep Learning Era: Masking Approach

Deep Learning Era: Architectural Evolution

Deep Learning Era: Time Domain Approach

Deep Learning Era: Generative Approach

Deep Learning Era: Vocoder-based Approach

Deep Learning Era: Foundational Model

Open Challenges & Conclusions

References

Speech Foundational Models for SE

Self-supervised (SSL) models — **wav2vec2.0**, **WavLM**, **HuBERT** — have successfully replaced standard speech representations across many tasks.

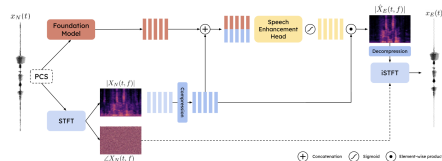
Early attempts [70, 71]: main issue was **missing fine-grained information** in SSL speech embeddings.

Recent improvements [72, 73] address this via:

- Cross-domain feature combination**

- Fine-tuning** of the SSL backbone

- Multiple losses** during training



LLMs for SE

LLMs represent speech as sequences of **discrete tokens** (mainly via **audio codecs**) rather than continuous waveforms.

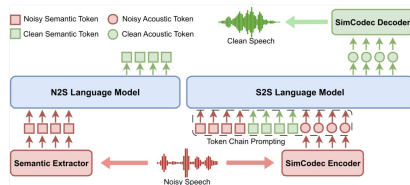
SE becomes a **sequence-to-sequence generation** problem, exploiting **long-range dependencies** and **linguistic priors** to reconstruct speech even under severe corruption.

SELM [74]: first such work.

Encoder: WavLM + k -means tokenizer

Decoder: de-tokenizer + **HiFi-GAN** vocoder

Transformer-based speech LLM performs
next-token prediction



GenSE [74]: Hierarchical extension that combines **semantic tokens** with **SimCodec** (a neural speech codec that extracts acoustic tokens via a quantizer and reconstructs speech via a decoder).

Followed by further work, e.g., [75].

Combining SSL & Vocoder for SE [76]

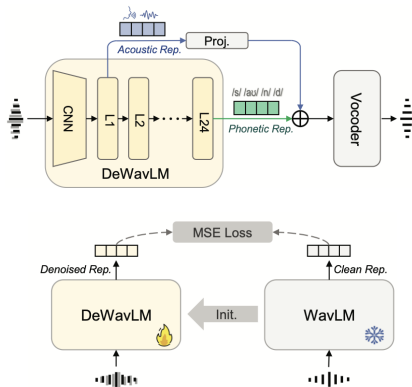
PASE (AAAI 2026): addresses a key issue in generative SE solutions — **linguistic and acoustic hallucinations**.

Proposed fix: combine two components:

A **denoising-expert WavLM**, obtained via **representation distillation**

Vocos, a cutting-edge neural vocoder with a strong **quality/efficiency trade-off**

Compared against: SpeechLLM-based SE, diffusion models, and discriminative solutions.



Outline

Introduction

Early Attempts

Deep Learning Era: Denoising Auto Encoders

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Deep Learning Era: Masking Approach

Deep Learning Era: Architectural Evolution

Deep Learning Era: Time Domain Approach

Deep Learning Era: Generative Approach

Deep Learning Era: Vocoder-based Approach

Deep Learning Era: Foundational Model

Open Challenges & Conclusions

References

Open Challenges

Computational efficiency

Many SE models remain too **resource-intensive** for latency-critical settings such as hearing aids and on-device assistants

Open problem: cut inference cost **without sacrificing** enhancement quality

Robustness in the wild

Real acoustic environments combine multiple, simultaneous degradations — **noise**, **reverberation**, and beyond

Models must generalize across this **diversity of conditions**, not just curated benchmarks

Evaluating faithfulness, not just quality

Current **intrusive and non-intrusive metrics** fail to catch **hallucinated content**

A cleaner-sounding output is not necessarily a **faithful** one — new metrics are needed

Conclusion

Mapping / masking: Effective at noise reduction, High faithfulness but point based estimation $\implies$ over-smoothing, or reduced perceptual quality.

Vocoder-, Diffusion-, and LLM-based models: High perceptual quality, but prone to **hallucinations** — inconsistencies in linguistic content or speaker identity

New metrics (WER, speaker similarity) are emerging to capture these failure modes

Open question

This shift calls for rethinking **evaluation**, **robustness**, and the very notion of “**clean**” **speech** — no longer just noise-free, but faithful to the original speaker and content.

Outline

Introduction

Early Attempts

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Deep Learning Era: Mapping Approach

Deep Learning Era: Masking Approach

Deep Learning Era: Architectural Evolution

Deep Learning Era: Time Domain Approach

Deep Learning Era: Generative Approach

Deep Learning Era: Vocoder-based Approach

Deep Learning Era: Foundational Model

Open Challenges & Conclusions

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